(a) Heterogeneous Fleet

▲ Scout Robots (Mobility)

• Speed: 2.0× nodes/tick
• Sensing Radius: 2 hops
• Comm Budget: 100 bits/tick
• Role: Recon & prior P(θ) update

■ Heavy Lifters (Payload)

• Speed: 1.0× nodes/tick
• Battery: 300 units (Move: 6.0)
• Sensing: 1 hop | Comm: 40 bits
• Role: High-value task fulfillment

(b) Adversary Policy Space (θ*)

θ1: Stochastic Jammer

• P(Xe = 1 | θ1) = P_base
• Random non-targeted blockage

θ2: Reactive Path Blocker

• P(Xe = 1 | θ2) = min(1.0, Pbase + αNe)
• Triggered within 1 hop of robot

θ3: Optimal Minimax

• argmax_e Σ ΔCost_r(e)
• Maximum bottleneck congestion

(c) Beliefs & KL Regularization

Bayesian Posterior Filter

P_r(θj | Xe) ∝ P(Xe | θj) P_r(θj)
• Private Prior: B_r = [P(θ1), P(θ2)]
• Inferred Teammate Mean: Q^(r)

KL-Regularized Bidding

Bid_r(t) = E_r[Cost] + λ D_KL(B_r || Q_r)
• Penalizes out-of-sync bids
• Stabilizes consensus under comm drop